Exam 3 Review

ENGR 0021

What Will Be on the Exam?

- Regression (interpretation, prediction, R^2)
- Hypothesis testing (z, t, χ^2, ANOVA)
- Choosing the correct statistical test
- P-values and conclusions
- Type I and Type II errors

Scatterplots & Correlation

From the exam:

- Estimate mean and standard deviation
- Determine sign of correlation

Key ideas:

- Mean = center of data
- SD = spread (about 68% within $\pm$ 1 SD, about 95% within 2 standard deviations)
- Correlation:
  - Positive $\rightarrow$ upward trend
  - Negative $\rightarrow$ downward trend
  - Near 0 $\rightarrow$ no linear relationship

Regression Interpretation

Model:

$$\hat{y} = -0.65 + 2.28x$$

What does the slope mean?

Intercept?

$\hat{y}_0 = -0.65$

R^2 Interpretation

$r^2 = R^2$

$$R^2 = 0.957$$

Meaning: how much variability in y explained by the model

High vs Low:

Common mistake:

- Thinking R^2 implies causation

Choosing the Right Test

Key question: What type of data?

- Proportion $\rightarrow$ z-test
- Mean (1 sample) $\rightarrow$ one-sample t
- Mean (2 samples) $\rightarrow$ two-sample t
- Counts/categories $\rightarrow$ χ^2
- Multiple means $\rightarrow$ ANOVA (F test)

Exam strategy:

- Identify parameter first
- Then match test

Hypothesis Testing Framework

1. State H_0 and H_1
2. Choose test
3. Compute test statistic
4. Find P-value
5. Compare to α

Decision rule:

$$\text{Reject } H_0 \text{ if } P < \alpha$$

P-Value Interpretation

Definition: not $P(H_0 \mid \text{data})$

- Probability of observing data this extreme assuming H_0 is true

Interpretation:

- Small P $\rightarrow$ evidence against H_0
- Large P $\rightarrow$ data consistent with H_0

Common mistake:

- P-value is NOT probability H_0 is true

Type I vs Type II Errors

Type I Error false positive $\alpha = .05$

- Reject H_0 when it is true

Type II Error false negative β

- Fail to reject H_0 when it is false

Exam tip:

- Always describe errors in context

Example: Proportion Test (Exam Review)

A company claims that 60% of its customers prefer its new product. In a random sample of 800 customers, 456 say they prefer the new product. You want to determine whether the data are consistent with the company's claim.

(a) Define the parameter and state the hypotheses.

$H_0: a = .6$

$H_1: a \neq .6$

(b) Which test should be used?

(A) One-sample z-test for a proportion

(B) Two-sample z-test

(C) One-sample t-test

(D) Chi-square test

z-test

$$\sigma = \sqrt{\frac{a(1-a)}{n}}$$

(c) One-sided or two-sided?

two-sided

(d) Compute the test statistic:

$$z = \frac{456/800 - .6}{SE(a)} = \frac{\frac{456}{800} - .6}{\sqrt{\frac{.6 \times .4}{800}}} = -1.73$$

(e) Approximate the P-value:

$$z = \frac{456 - 480}{\sqrt{n a(1-a)}} = \frac{456 - 480}{\sqrt{800 \times .6 \times .4}} = -1.73$$

$$P = 2P(z > 1.73) = .0836 \;;$$

$$P > \alpha \;;$$ do not reject H_0

Chi-Square Test (Variance)

$$\chi^2 = \frac{(n-1)s^2}{\sigma_0^2}$$

Key ideas:

- Used for variance
- Requires normality
- Uses degrees of freedom $n - 1$

goodness-of-fit:

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

ANOVA (F-Test)

Purpose:

- Compare more than 2 means

Hypotheses:

$$H_0 : \mu_1 = \mu_2 = \cdots$$

$$H_1 : \text{Not all equal}$$

Key idea:

$$F = \frac{\text{between variation}}{\text{within variation}}$$

ANOVA Table

		Df	Sum Sq	Mean Sq	F value	Pr(>F)
(**G**roup)	depth	2	16.96	8.48	6.13	0.0063
(**E**rror)	Residuals	27	37.33	1.38		
	Total	29	54.29			

$df_G = k-1$

$df_T = n-1$

$SS_{Total} = \sum (y_i - \bar{y})^2$

$SS_{Group} = \sum n_j (\bar{y}_j - \bar{y})^2$

$SS_{ERROR} = \sum (y_{ij} - \bar{y}_j)^2$

$p \stackrel{*}{=} P(F_{2,27} > 6.13)$

Regression Results

	coef	std err	t	$P > \|t\|$	[0.025	0.975]
const	9.2076	9.300	0.990	0.332	-9.946	28.361
biological	0.9014	0.096	9.358	0.000	0.703	1.100

$$\hat{y} = 9.2076 + .9014x$$

$$SE(b_0) = 9.3, \quad SE(b_1) = .096$$

$$t = \frac{b_j}{SE(b_j)}$$

$$P = P(T_{df} > t)$$

$$b_j \pm t_{\alpha/2, n-2} SE(b_j)$$

$$t_{\alpha/2, n-2} \approx 2$$

Confidence Intervals for Slope

$$b_1 \pm t_{\alpha/2,\, n-2} \cdot SE(b_1)$$

Interpretation:

- Range of plausible values for slope

Connection to hypothesis test:

- If 0 NOT in CI $\rightarrow$ significant

Final Tips

- Always define variables clearly
- Write hypotheses properly
- Interpret results in context
- Watch for:
  - one-tailed vs two-tailed
  - correct test choice
  - units and meaning

Most common mistake: choosing the wrong test

Practice Question

Which test would you use?

- Compare 3 machine settings
- Survey preference (yes/no vs gender)
- Test average vs known value

ANOVA

χ^2 for independence

two-sample t-test

one-sample t-test or z-test

Questions?

Good luck on the exam!